

17 An object of weight 3.0 N is moved vertically at a constant velocity v by an electric motor.

The electric motor has an efficiency of 25% .

The object travels a vertical distance of 1.2 m in 3.0 s .

Assume that the useful output energy of the motor is equal to the increase in gravitational potential energy of the object.

What is the electric power supplied to the motor?

A 0.3 W

B 1.2 W

C 4.8 W

D 30 W